

ODE Coefficient Synthesis

The coefficient lattice and its staircase

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2026-10-08

Abstract

Dimensional constraints give a table of admissible ODE coefficients. Minimizing absolute integer exponents selects a staircase with recurrence $A_{k+2} = LC A_k$. Finite sums assemble higher-order equations, and $D = i\omega$ gives their phasor form.

PDF created: 2026-10-08 13:07:40 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-ode-coefficient-synthesis>

1 Deriving the coefficient family

Start with $aD^2y + bDy + cy = f$ from [the ODE template](#), where $D = d/dt$ and $a, b, c > 0$ are constants. With independent dimensions $U = [c]$ and time T , equal term dimensions require

$$[a] = UT^2, \quad [b] = UT, \quad [c] = U.$$

A coefficient of $D^k y$ must have dimension UT^k . For a monomial $a^p b^q c^r$, this means

$$p + q + r = 1, \quad 2p + q = k.$$

Solving gives the family, up to a dimensionless multiplier,

$$A_{r,k}(a, b, c) = a^{k-1+r} b^{2-k-2r} c^r. \quad (1)$$

We choose integer r, k : the powers may be positive, zero or negative. Dimensions alone also allow real r ; the integer restriction defines the Laurent-monomial class used here.

With $(a, b, c) = (L, R, 1/C)$,

$$A_{r,k} = L^{k-1+r} R^{2-k-2r} C^{-r}. \quad (2)$$

For example, $k = 3$ and $r = 0, -1, -2$ give L^2/R , LRC and $R^3 C^2$. Dimensions leave one exponent free; selecting a coefficient needs another rule.

2 The coefficient table

Table 1 lists (2); boxes mark the staircase selected in Section 3. ODEs use $k \geq 0$. Negative columns extend the dimension rule to repeated integrals, with time factors T^{-k} .

$r \backslash k$	-4	-3	-2	-1	0
3	$\frac{1}{L^2 C^3}$	$\frac{1}{LRC^3}$	$\frac{1}{R^2 C^3}$	$\frac{L}{R^3 C^3}$	$\frac{L^2}{R^4 C^3}$
2	$\frac{R^2}{L^3 C^2}$	$\frac{R}{L^2 C^2}$	$\frac{1}{LC^2}$	$\frac{1}{RC^2}$	$\frac{L}{R^2 C^2}$
1	$\frac{R^4}{L^4 C}$	$\frac{R^3}{L^3 C}$	$\frac{R^2}{L^2 C}$	$\frac{R}{LC}$	$\frac{1}{C}$
0	$\frac{R^6}{L^5}$	$\frac{R^5}{L^4}$	$\frac{R^4}{L^3}$	$\frac{R^3}{L^2}$	$\frac{R^2}{L}$
-1	$\frac{R^8 C}{L^6}$	$\frac{R^7 C}{L^5}$	$\frac{R^6 C}{L^4}$	$\frac{R^5 C}{L^3}$	$\frac{R^4 C}{L^2}$
-2	$\frac{R^{10} C^2}{L^7}$	$\frac{R^9 C^2}{L^6}$	$\frac{R^8 C^2}{L^5}$	$\frac{R^7 C^2}{L^4}$	$\frac{R^6 C^2}{L^3}$
-3	$\frac{R^{12} C^3}{L^8}$	$\frac{R^{11} C^3}{L^7}$	$\frac{R^{10} C^3}{L^6}$	$\frac{R^9 C^3}{L^5}$	$\frac{R^8 C^3}{L^4}$

$r \backslash k$	1	2	3	4	5	6
3	$\frac{L^3}{R^5 C^3}$	$\frac{L^4}{R^6 C^3}$	$\frac{L^5}{R^7 C^3}$	$\frac{L^6}{R^8 C^3}$	$\frac{L^7}{R^9 C^3}$	$\frac{L^8}{R^{10} C^3}$
2	$\frac{L^2}{R^3 C^2}$	$\frac{L^3}{R^4 C^2}$	$\frac{L^4}{R^5 C^2}$	$\frac{L^5}{R^6 C^2}$	$\frac{L^6}{R^7 C^2}$	$\frac{L^7}{R^8 C^2}$
1	$\frac{L}{RC}$	$\frac{L^2}{R^2 C}$	$\frac{L^3}{R^3 C}$	$\frac{L^4}{R^4 C}$	$\frac{L^5}{R^5 C}$	$\frac{L^6}{R^6 C}$
0	$\frac{R}{L}$	$\frac{L}{R}$	$\frac{L^2}{R}$	$\frac{L^3}{R^2}$	$\frac{L^4}{R^3}$	$\frac{L^5}{R^4}$
-1	$\frac{R^3 C}{L}$	$R^2 C$	$\frac{LRC}{L}$	$\frac{L^2 C}{L}$	$\frac{L^3 C}{R}$	$\frac{L^4 C}{R^2}$
-2	$\frac{R^5 C^2}{L^2}$	$\frac{R^4 C^2}{L}$	$R^3 C^2$	$LR^2 C^2$	$\frac{L^2 RC^2}{L}$	$\frac{L^3 C^2}{L}$
-3	$\frac{R^7 C^3}{L^3}$	$\frac{R^6 C^3}{L^2}$	$\frac{R^5 C^3}{L}$	$R^4 C^3$	$LR^3 C^3$	$L^2 R^2 C^3$

Table 1. The coefficient table for $(a, b, c) = (L, R, 1/C)$, with $r = 3, \dots, -3$ and $k = -4, \dots, 6$. Blue boxes select $r_k = \lfloor (2 - k)/2 \rfloor$.

Two ratios generate the table:

$$A_{r,k+1} = \frac{L}{R} A_{r,k}, \quad A_{r-1,k} = \frac{R^2 C}{L} A_{r,k}. \quad (3)$$

A step right contributes a time factor; a step down is dimensionless.

3 Selecting the staircase

Minimize the sum of absolute exponents in the declared basis (L, R, C) :

$$F_k(r) = |k - 1 + r| + |2 - k - 2r| + |r|.$$

Break ties by choosing a nonnegative exponent of R . Then

$$r_k = \left\lfloor \frac{2 - k}{2} \right\rfloor = 1 - \left\lceil \frac{k}{2} \right\rceil. \quad (4)$$

Indeed, the triangle inequality gives

$$F_k(r) \geq |k - 1| + |2 - k - 2r|.$$

The last term is at least 0 for even k and 1 for odd k . The chosen r_k attains this bound and gives those same nonnegative exponents of R , satisfying the minimum and tie rule.

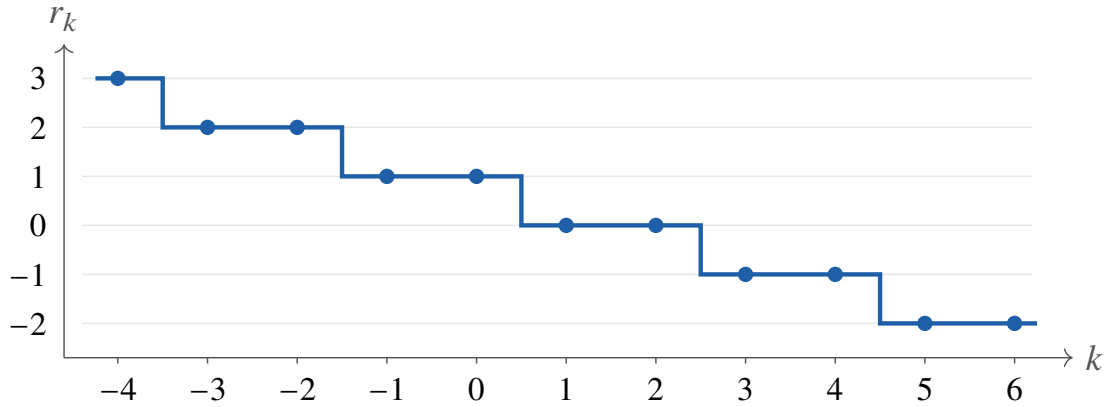


Figure 1. The selected staircase $r = r_k$. Each pair of columns shares one row; the next pair is one row lower.

Set $A_k = A_{r_k, k}$. Substitution gives

$$A_{2n} = L^n C^{n-1}, \quad A_{2n+1} = R(LC)^n, \quad n \in \mathbb{Z}, \quad (5)$$

and therefore

$$A_{k+2} = LC A_k, \quad A_0 = \frac{1}{C}, \quad A_1 = R. \quad (6)$$

The even and odd coefficients are two geometric sequences. Together they begin $1/C, R, L, LRC, L^2C, L^2RC^2, L^3C^2$.

4 Assembling an ODE

For integer $N \geq 1$, define

$$P_N(D)y = f, \quad P_N(D) = \sum_{k=0}^N A_k D^k. \quad (7)$$

In the original symbols,

$$A_{2n} = a^n c^{1-n}, \quad A_{2n+1} = b(a/c)^n, \quad A_{k+2} = \frac{a}{c} A_k.$$

The cubic factors as

$$\frac{ab}{c} D^3 + aD^2 + bD + c = (bD + c) \left(1 + \frac{a}{c} D^2 \right). \quad (8)$$

Its coefficient $ab/c = LRC$ is used in the signed-work calculation of *Energy Ledgers for Forced Harmonic ODEs*, Sections 2–3. Pairing adjacent orders gives

$$P_{2m+1}(D) = (c + bD) \sum_{j=0}^m \left(\frac{a}{c} D^2 \right)^j, \quad m \geq 0. \quad (9)$$

For $m \geq 1$, obtain P_{2m} by adding $a^m c^{1-m} D^{2m}$ to P_{2m-1} . These finite identities require no convergence assumption.

4.1 Other selections

Let $S = b^2/a$, $\tau = a/b$ and $\rho = b^2/(ac)$, with dimensions $U, T, 1$, respectively. Then

$$A_{r,k} = S \tau^k \rho^{-r}, \quad B_k = \sum_{r \in \mathcal{R}_k} w_{r,k} A_{r,k} = S \tau^k \sum_{r \in \mathcal{R}_k} w_{r,k} \rho^{-r}, \quad (10)$$

where each $\mathcal{R}_k \subset \mathbb{Z}$ is finite and $w_{r,k}$ are real dimensionless constants. The resulting $Q(D)y = f$, $Q(s) = \sum_{k=0}^N B_k s^k$, has order N if $B_N \neq 0$. The staircase takes $\mathcal{R}_k = \{r_k\}$ and $w_{r_k,k} = 1$; weighted sums retain other dimensionally admissible choices.

5 Phasor form

Use $y(t) = \text{Re}(\hat{y}e^{i\omega t})$, $i^2 = -1$, $\omega > 0$. Substituting $D = i\omega$ gives

$$\boxed{Q(i\omega)\hat{y} = \hat{f}, \quad Q(i\omega) = \sum_{k=0}^N B_k (i\omega)^k.} \quad (11)$$

Division requires $Q(i\omega) \neq 0$; a nonzero homogeneous phasor requires $Q(i\omega) = 0$. These are harmonic components, with initial data for the full solution specified separately.

5.1 Frequency factors

Multiply column k of the table by $(i\omega)^k$:

$$\Phi_{r,k} = A_{r,k} (i\omega)^k = S \rho^{-r} (i\omega \tau)^k. \quad (12)$$

The factors i^k cycle through $1, i, -1, -i$. For negative k , this gives the harmonic antiderivative; integration constants are separate. On the staircase,

$$\Phi_{2n} = \frac{1}{C} (-\omega^2 LC)^n, \quad \Phi_{2n+1} = i\omega R (-\omega^2 LC)^n, \quad n \in \mathbb{Z}. \quad (13)$$

Thus $\Phi_{k+2} = -\omega^2 LC \Phi_k$: every two orders reverse the sign and multiply the magnitude by $\omega^2 LC$. Pairing terms gives

$$P_{2m+1}(i\omega) = \left(\frac{1}{C} + i\omega R \right) \sum_{n=0}^m (-\omega^2 LC)^n, \quad m \geq 0. \quad (14)$$

In particular, $P_3(i\omega) = (1/C + i\omega R)(1 - \omega^2 LC)$. The finite sum is valid at every $\omega > 0$.

5.2 Using the derivative as terminal variable

For $u = Dy$, $\hat{u} = i\omega \hat{y}$, so

$$H(\omega) = \frac{Q(i\omega)}{i\omega} = \sum_{k=0}^N B_k (i\omega)^{k-1}, \quad \hat{f} = H(\omega) \hat{u}. \quad (15)$$

The exponent is k for y and $k - 1$ for Dy . [Interconnecting Series and Parallel ODEs](#), Section 3, combines these terminal ratios through common-variable and sum constraints.

References

1. H. Nilre and B. C. Herlin (2026). [The ODE Template and Its Domain Equivalents](#).
2. H. Nilre and B. C. Herlin (2026). [Interconnecting Series and Parallel ODEs](#).
3. H. Nilre and B. C. Herlin (2026). [Energy Ledgers for Forced Harmonic ODEs](#).